KOMA

Izpiti: 25.1 (21.2)
4.2

Kolokviji:
1) torek 24.11 ob 17.30
2) torek 12.1 ob 17.30

1) Naj bo k polje in $k[V]$ $k[X]$-algebra podana s $k[X] \to k[V]$

$X \mapsto V^2$

$\sum \mu_i X^i \mapsto \sum \mu_i V^{2i}$

$k[V]$ je kolobar in $k[X]$-modul

$\lambda(x \cdot y) = (\lambda x) y = x(\lambda y)$

a) pokaži da je $k[V]$ prosti modul $k[X]$ in $k[V] \underset{\text{vig}}{\cong} \frac{k[X][T]}{(T^2 - X)}$

b) Pokaži $B := k[V] \otimes_{k[V]} k[V] \overset{\text{kol}}{\cong} \frac{k[V,W]}{(V^2 - W^2)}$

c) Če $\operatorname{char} k \neq 2$ pokaži: da B ni cel kolobar, ima nima nilpotenta in $B \cong \{(f,g) \in k[V] \times k[V] ; f(0) = g(0)\}$

d) Naj bo $\operatorname{char} k = 2$. Poišči nilpotent (neničelni) v B

a) $\lambda_0 + \lambda_1 v + \dots + \lambda_n v^n$

$x \cdot 1 = v^2$

$x \cdot v = v^3$

$x \cdot v^n = v^{n+2}$

$(1+x) \cdot 1 = 1 + v^2$

$\{1, v\}$

$\lambda_0 + \lambda_1 v + \dots \lambda_n v^n =$

$(\lambda_0 + \lambda_2 v^2 + \dots) + v(\lambda_1 + \lambda_3 v^2 + \dots) =$

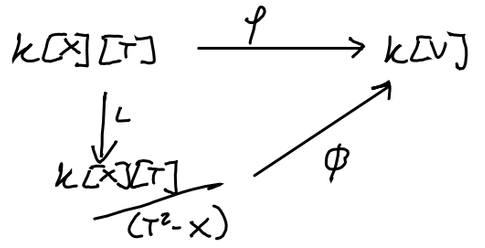
$= (\lambda_0 + \lambda_2 x + \dots) \cdot 1 + (\lambda_1 + \lambda_3 x + \dots) v$

$$k[v] \simeq \frac{k[x][T]}{(T^2 - x)}$$

$a_0 + a_1 v \longmapsto a_0 + a_1 T$

Ali to ohranja množenje? ...

2. Način



$\varphi: \quad T \longmapsto v$

$\quad\quad x \longmapsto v^2$

ohranja strukturo modula

$\sum_i \left(\left(\sum_j \mu x^j \right) v_i T^j \right) \longmapsto \left(\sum \mu v^{2j} \right) \left(\sum v_i v^j \right)$

$= \sum_{i,j} \mu_i x^j v_i T^j \longmapsto \sum \mu_i v_i v^{2j} w^j$ " ✓

Kaj je jedro?

$\ker \varphi = ?$

$(T^2 - x^k) \longmapsto v^2 - v^2 = 0$ to je v jedru

$\sum v_{jk} x^j T^k \longmapsto \sum v_{jk} x^j v^k =$

$= p(x) \cdot 1 + q(x) v$

$= \underbrace{\sum \lambda_i v^{2i}}_{=0} + \underbrace{\sum \gamma_j v^{2j+1}}_{=0}$

b)

$$k[V]\otimes \frac{k[X][T]}{(T-X)} \cong \frac{(k[V]\otimes k[X])(T)}{(T^2-X)} \cong$$

$$\cong \frac{k[V][T]}{(T^2-X)} \cong k[V]\frac{[T]}{(T^2-V)}$$

2. način

$$\varphi: k[V]\otimes k[V] \longrightarrow \frac{k[V,W]}{(V^2-W^2)}$$

$$\sum \mu_i (\rho_i(V)\otimes \xi_i(W)) \longmapsto \sum \mu_i \rho_i(V)\xi_i(W)$$

Univerzalna lastnost za algebre

$$\begin{array}{ccc} A & \xrightarrow{\hom_\varphi} & C \\ B & \xrightarrow{\hom_\psi} & \end{array} \quad \rightsquigarrow \quad A\otimes B \longrightarrow C$$

$$a\otimes b \longmapsto \varphi(a)\psi(b)$$

+ pogoj komutativnosti

Preslikavi sta identiteti

$$k[V] \longmapsto \frac{k[V,W]}{(V^2-W^2)}$$

$$V \longmapsto V$$

$$V \longmapsto W$$

To je preslikava v eno smer

Druga smer:

$$\Psi\ k[V,W] \longrightarrow k[V]\otimes k[V]$$

$$\sum_{ij} \lambda_{ij} V^i W^j \longmapsto \lambda_{ij}(V^i\otimes W^j)$$

$$V \longmapsto (V\otimes 1)$$

$$W \longmapsto (1\otimes W)$$

Po univerzalni lastnosti je to homomorfizem

$(V^2-W^2) \in \ker \Psi$

$$(V^2\otimes 1) - (1\otimes W^2) = ((X-1)-(X-1))(1\otimes 1) = 0$$

$\Rightarrow$ to je homomorfizem

$$\frac{k[V,W]}{(V^2-W^2)} \longrightarrow k[V]\otimes k[V]$$

Ali je to izomorfizem?

$\Phi\circ\overline{\Psi} = id$

Vstavimo v in w (če je na generatorjih id je povsod id

$$(\Phi\circ\overline{\Psi})(V) = \Phi(V\otimes 1) = V$$

$$\overline{\Psi}(\Phi(f\otimes g)) = \Psi(f(V)\cdot g(W)) = (f\otimes g)$$

c)

$$\mathcal{B} = k[V] \otimes k[V] = \frac{k[v,w]}{(v^2-w^2)}$$

$$\mathcal{B} \cong \{(f,g);\ h[V] \times h[V];\ f(0) = g(0)\}$$



B_cel

- komutativnost ✓
- obrnljivost ?

  nima deliteljev ničla

$$\frac{k[v,w]}{(v-w)(v+w)} \longrightarrow$$

crt: I, J ideala, ki sta si tuja v algebri A

$I + J = A$

Potem $A/_{IJ} \cong A/_I \times A/_J$

Al: sta tuja

$$I + J = p(x)(v-w) + g(x)(v+w) =$$

$$= (p(x)+g(x))(v) + (p(x)-g(x))w = (v,w)$$

Boj splošno

$$A/_{I \cap J} \cong \{(f,g) \in A/_I \times A/_J;\ f \equiv g \bmod (I+J)\}$$

$$A \longrightarrow C \qquad \text{surjektivnost}$$

$$a \mapsto (a \bmod I, a \bmod J)$$

$I \cap J = (v^2 - w^2)$

$\supseteq$ ✓

$\subseteq$ $f \in I \cap J \Rightarrow (v-w) \mid f$ in $(v+w) \mid f$

ker sta nerazcepna in nessociirana

$\underset{UFD}{\Rightarrow}$ bo $(v-w)(v+w) \mid f$

$$\Rightarrow \cong \{(f,g) \in \frac{k[v,w]}{(v-w)} \times \frac{k[v,w]}{(v+w)}\}$$

$$\frac{k[u,v]}{(v \pm w)} \cong k[y]$$

$f(v,w) \longmapsto f(y, \mp y)$ ta je surjektivna

$f(0) = g(0)$

$\lambda_0 = \gamma_0 \bmod (v,w)$

⋮

$R = \mathcal{P}(X)$ enota

(2.) $A + B = (A - B) \cup (B - A)$ $\emptyset$

$A \cdot B = A \cap B$ $\mathcal{U}$

a) $I \circ R \Leftrightarrow$ zapite za podmnožice in končne unije

$I \subseteq R$

$I \neq \emptyset$

($\Leftarrow$)

$A, B \in I$ –

$A + B = (A - B) \cup (B - A) \Rightarrow$ ✓

($A - B \subseteq A$, $B - A \subseteq B$)

$A \cdot B = A \cap B \subseteq A$

($\Rightarrow$) Naj bo $A \in I$ $B \subseteq A$

$\underline{B = A \cap B = A \cdot B \in I}$

$A, B \in I$

$A \cup B = \underbrace{(A - B) \cap (B - A)}_{\in I = C} \cup \underbrace{(A \cap B)}_{\in I} =$

$C \cup (A \cap B) =$

$(C - (A \cap B)) \cup ((A \cap B) - C) = A \cap B + C \in I$

b) Dokaži $\langle A \rangle$ in $\langle A, B \rangle$

$$\mathcal{R}\cdot A = \{ B \cap A ; B \in \mathcal{R} \} = \mathcal{P}(A)$$

$\langle A, B \rangle$

$A \cup B \in \langle A, B \rangle \Rightarrow \langle A, B \rangle \subseteq \mathcal{P}(A \cup B)$

- $\mathcal{P}(A \cup B) \subseteq \langle A, B \rangle$

$C \in \mathcal{P}(A \cup B) = (\underbrace{C \cap A}_{\in I}) \cup (\underbrace{C \cap B}_{\in I}) =$

$= ((C \cap A) - (C \cap B)) \cup (C \cap B) - (C \cap A)) \cup (C \cap A \cap B)$

$\langle A, B \rangle \subseteq \mathcal{P}(A \cup B)$

$(C \cap A) + (C \cap B) \subseteq A \cup B \rightsquigarrow C \in \mathcal{P}(A \cup B)$

$\langle A_1, \ldots A_n \rangle = \langle A_1 \cup A_2, A_3, A_4 \ldots A_n \rangle = \ldots = \langle A_1 \cup A_2 \cup A_3 \ldots \rangle$

c) X neskončna

$\mathcal{I} = \{ A \subseteq X$ je končna $\}$ je ideal ki ni končno generiran

Recimo da je končno generiran.
Potem $\mathcal{I} = \mathcal{P}(A_1 \cup A_2 \cup \ldots \cup A_n) = \mathcal{P}(A)$

A je končna, X pa ima neskončno enojce

$\Downarrow$ $\mathcal{P}(A)$ je končna)

$\Downarrow$ $\mathcal{I}$ je neskončen

Ali je to ideal po točki a) ja

Boolov kolobar ... vsi so idempotentni
$A \cdot A = A$

Vsi Boolov kolobar $\cong P(S)$ za neko končno S
končen

$a \cdot a = a$ $R = \{a_1 \dots a_n\}$ je boolov

~~Velja: $A \cap A = A \Rightarrow R = P(X)$ je boolov~~

$P(S)$ ima n elementov
želimo $n = 2^k$
$n = 2^k$

$\exists\, \forall a_i \in A.\ A \cong A/(1-a) \times A/a$
$R \cong R \cdot e \times R(1-e) \cong$
obe sta boolova s končno elementi
indukcija po moči $2^{n_1} \cdot 2^{n_2} = 2^{n_1+n_2}$ ✓
$n = 2 \dots n = 2^\Phi$ ✓

$R \cong \mathbb{Z}_2 \times \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$

Imamo S s k elementi. $S = [k]$
$P(S)$

$R = \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2 \cong S(k)$ je končen boolov kolobar (je izomorfen $\mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$)

$\mathbb{Z}_2^{\mathbb{N}}$
$R = \{a \in \mathbb{Z}_2^{\mathbb{N}}$ končno mnogo a_i neničelnih$\}$ neskončno mnogo enk
$R = \bigoplus_{\mathbb{N}} \mathbb{Z}_2$

↑ števno neskončen $P(S)$ je ali končen ali nešteven neskončen

$A[[X]] = \{ f = \sum_{n \geq 0} a_n X^n \quad a_n \in A \}$

a) f obrnljiv v $A[[X]] \Leftrightarrow a_0$ obrnljiv v A

Ne bo blo

b) Naj bo K polje

$I \trianglelefteq K[[X]] \quad I \neq 0 \Rightarrow I = \langle X^n \rangle$

Naj bo I neničeln ideal

$$\sum a_n X^n \cdot \sum b_k X^k = \sum_n \sum_{k=0}^{n} a_{n-k} b_k X^n$$

$\langle X^n \rangle = \{ p(x) X^n \}$

x je najmanjši n da $a_n \neq 0$ za

$\sum a_k X^k$ lahko zapišemo kot $X^n \sum_{k \geq n} a_{k-n} X^k$

$\Rightarrow I \subseteq \langle X^n \rangle$

$X^n \in I$

$X^n \sum_{k=0}^{\infty} a_{k+n} X^k \qquad a_{k+n} \neq 0$

ker a_n ni 0 je dobljen

c)

$\mathrm{char}\, k \neq 2 \quad 1+x = f^2 \quad f \in k[[x]]$

$$\left(\sum a_n x^n\right)^2 = \sum_{n=0}^{\infty} \sum_{k=0}^{n} a_{n-k} b_k x^n = 1 + x$$

$$(1+x)^{\frac{1}{2}} = \sum a_0 x^n \quad \text{taylorjeva vrsta}$$

$$= \sum \binom{\frac{1}{2}}{k} x^k$$

$$\binom{\frac{1}{2}}{k} = \frac{\overbrace{\frac{1}{2} \cdot \left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\ldots\left(\frac{1}{2}-k+1\right)}^{k}}{k!} = \ldots$$

$$= \frac{2^k}{k!}(-1)(-3)(-5)\ldots(1-2k) =$$

$$\frac{(2k-1)!!\,(-2)^k}{k!}$$

Kaj pa če $\mathrm{char}\, k \mid k$

$$a_0 b_0 = 1$$

$$a_0 b_1 + a_1 b_0 = 1$$

oziroma je 0 če $a_0 \neq 0$ je to vredu

$$a_0^2 = 1 \qquad \text{Recimo da } a_0 = 1$$

$$2 a_0 a_1 = 1 \qquad \Rightarrow a_1 = \frac{1}{2}$$

$$a_1^2 + 2 a_2 a_0 = 0 \qquad \Rightarrow a_2 = -\frac{1}{2^3}$$

$$2 a_3 a_0 + 2 a_1 a_2 \Rightarrow a_3 = -\frac{a_1^3 a_2}{a_0} = \frac{1}{2}$$

Kaj če char 2 ?

$$x(a_1 a_0 \cdot 2) = 0 \neq 1$$

③ e, e' idempotenta v R

$(e)^2 = (e)$

$(e) = Re$

$(e)^2 = (Re)(Re) =$

$= \{ae \cdot be\} = \{ab e^2\} = \{ab e\} \in (e)$

$(e) \subseteq (e)^2$

obvs. $e = e \cdot e$

$(a)^2 = (a) \Rightarrow a = (f)$ za nek f

$(e) = (e') \Rightarrow e = e'$

$Ra = Ra \cdot Ra = \{ba \cdot ca\} = \{bc \cdot a^2\}$

$(a)^2 = (a^2)$

$(a) = (a^2)$

$Ra = Ra^2$

$ba = ca^2$

$ka^2 = a \quad / k$

$(ka)^2 = ka \Rightarrow ka$ je idempotent

$(ka)R = (R \cdot k) \cdot a \subseteq R \cdot a$

$a = ka^2 \Rightarrow a \in R \cdot k \cdot a \Rightarrow Ra \subseteq R \cdot ka$

Rezimo da

$(e) = (e')$

$Re = Re'$

$e' = ke$

$e = ce'$

$e = cke$

a) $\mathbb{F}_p[x] = \mathbb{Z}_p[x]$

$\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$

$\varphi \; \mathbb{Z}[x] \longrightarrow \mathbb{F}_p[x]$
$\sum a_i x^i \mapsto \sum (a_i \bmod p)\, x^i$

$\ker \varphi = \{p \cdot f(x);\ f(x) \in \mathbb{Z}\} = p \cdot \mathbb{Z}[x] = (p)$

$\mathbb{F}_p[x] \cong \frac{\mathbb{Z}[x]}{p\mathbb{Z}[x]}$

$\mathbb{Z}[x] \xrightarrow{\psi} \mathbb{Z}[i]$
$x \longmapsto i$
$\sum a_i x^i = \sum a_j i^j$ To je eval

$\ker \varphi = \{f(x);\ f(i) = 0\}$
$= (x^2+1)$

$f(x) = p(x^4) + x g(x^4) + x^2 r(x^4) + x^3 s(x^4)$
$f(1) = p(1) + i\, g(1) - r(1) - i\, s(1)$

$\mathbb{Z}[x]/(x^2+1) \xrightarrow{\psi} \mathbb{Z}[i]$

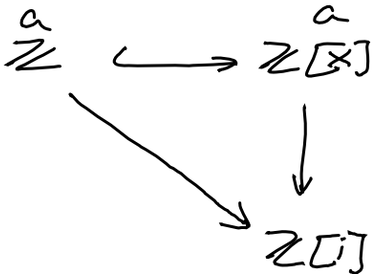
če je $\ker \psi = (x^2+1)$
potem je izo izbrano

$f = \sum a_j x^j + (x^2+1)$

$f = \sum b_0 + b_1 x + (x^2+1)$

$\psi(f) = \sum b_0 + b_1 i$
$= 0 \iff b_0 = 0 = b_1 = 0 \iff f = 0$

$x^3 = x^2 \cdot x =$
$= ((x^2+1) - 1) \cdot x =$
$0 - x = -x$



$$\mathbb{F}_p/(x^2+1) \cong \mathbb{Z}[i]/(p)$$

$$\mathbb{F}_p[x]/(x^2+1) \cong \mathbb{Z}[x]/(p)/(x^2+1) \cong \mathbb{Z}[x]/(p, x^2+1) \cong$$

$$\cong \mathbb{Z}[x]/(x^2+1)/(p) \cong \mathbb{Z}[i]/(p)$$

Kdaj je -1 kvadrat v $\mathbb{Z}_p$?

$$-1 = a^2 \Rightarrow a^2 = p-1 \bmod p$$

-1 ni kvadrat v $\mathbb{F}_p \Leftrightarrow$

$$\mathbb{F}_p^\times = \mathbb{Z}/(p-1) = \langle \zeta \rangle = \{1, \zeta, \dots \zeta^{p-2}\}$$

$$x^2 = -1 \quad \zeta^{k2} = -1 = \zeta^{\frac{p-1}{2}}$$

$$2k = \frac{p-1}{2} \bmod p-1$$

$$k = \frac{p-1}{4} \quad \vee \quad k = -\frac{p-1}{4}$$

$$p-1 = 0 \bmod 4$$

$$\Rightarrow (\zeta^{\frac{p-1}{4}})^2 = -1$$

Za kolobarje določi maksimalne ideale in enote
b) kolik idealov ki niso max imajo

$\mathbb{Z}/(2)[X] = \mathbb{Z}_2[X]$

f - nerazcepen

$X+1, X, X^2+X+1$ so edini nerazcepni

enote: 1 $\quad \mathbb{Z}/(2)[X]^\times = \{1\}$

M max $\Rightarrow$ M prime

$(f) = (gh) \Rightarrow \dfrac{\mathbb{Z}/(2)[X]}{(h)}$ cel? NE

$gh = 0$

$\mathbb{R}[[X]]$

units: tisti kjer je konstantni člen obrnljiv

vsi ideali: (0), $\langle X^n \rangle$ $n \in \mathbb{N}_0$ (projekcije vaje)

maximalni: $\langle X \rangle$

praideali: $\langle 0 \rangle, \langle X \rangle$

$\mathbb{Z}[X]$:

$\mathbb{Z}[X]^\times$ = prosti člen je 1 ali -1

maximalni: $\langle X, p \rangle$ $p \in \mathbb{P}$

I nek max ideal $X \in I$

$I + \langle X \rangle \neq \mathbb{Z}[X]$

Recimo da $I + (X) = \mathbb{Z}[X]$

$1 = f + X \cdot g(X)$

f ima konstantni člen $1 \Rightarrow f$ je obrnljiv

$\Rightarrow I = \mathbb{Z}[X]$

Torej I ali vsebuje $\langle X \rangle$ ali pa je od ko se x dodela

praideali: $\langle 0 \rangle$, maksimalni: $\langle X \rangle, \langle p \rangle$

$(X^2 - 6), \ldots$ neskončno

npr $(X-6)$ ni praideal

$\mathbb{Z}_{(2)} = \{\frac{a}{b} \in \mathbb{Q} \mid b \text{ lih}\}$

~~ideali~~ $\{\frac{a}{b}; a \text{ sod. } b \text{ lih}\} = \langle 2 \rangle$

ende: $\{\frac{a}{b}; a, b \text{ lih}\}$

torej je $\langle 2 \rangle$ edini maximalni

praideali: $\langle a \rangle, \langle 2 \rangle, \langle 2^a \rangle$

vsi ideali so vsebovani v

$\mathbb{Z} \times \mathbb{Z}$: vsi ideali $n\mathbb{Z} \times m\mathbb{Z}$ $n, m \in \mathbb{N}_0$

maksimalni $p\mathbb{Z} \times \mathbb{Z}$, $\mathbb{Z} \times p\mathbb{Z}$